

Baseline

1. The baseline uses the pretrained [all-MiniLM-L6-v2](#) sentence-transformer without any training.
2. The PDFs are cleaned (headers, footers, page numbers and hyphenation removed) and split into chunks of about 100 words with one sentence of overlap.
3. Each chunk is encoded with the model at a maximum sequence length of 128 tokens and L2-normalized, giving 384-dimensional vectors. To match the dimensionality of the custom model, these are reduced to 64 dimensions with PCA, fitted on the training-document chunks only.
4. The embeddings are then clustered with KMeans (seed 42), where k is chosen by a silhouette sweep from 2 to 15.
5. Clusters are evaluated with internal metrics (silhouette, Davies-Bouldin, Calinski-Harabasz) and with ARI and NMI against the source document.

Domain specific embedder

1. The custom embedder is the same MiniLM encoder with a linear 384-to-64 projection head, L2-normalized.
2. The projection is initialized from PCA of the pretrained embeddings, so the model starts out identical to the reduced baseline and any difference comes from training.
3. It is fine-tuned, with the whole encoder trainable, on chunks from the 3 training documents only; the 2 held-out documents are never seen.
4. Training is self-supervised. For every chunk, two positives are sampled from the same document: 75% from nearby chunks (within ± 3 positions) and 25% from anywhere in the document.
5. The loss is a symmetric InfoNCE contrastive loss with a scale of 20, using the other pairs in the batch as negatives.
6. Chunks from the same document are masked out of the negatives, since they would otherwise be false negatives.
7. I trained for 3 epochs with batch size 64, using AdamW with learning rates of $2e-5$ for the encoder and $1e-4$ for the projection, 10% linear warmup followed by linear decay, gradient clipping at 1.0 and fp16 mixed precision.
8. Afterwards, all chunks are embedded and clustered with the same algorithm, k and seed as the baseline, so the two models are compared under identical conditions.

Results

Overview (V1): The results show a case of overfitting, since we have only 3 docs for training, the embedding model learns to separate the training docs but overfits the test docs.

AIR and NMI confusion (V1):

- The high values of ARI initially gives an intuition that the embedding model improved the clustering and aligned it more with actual documents.
- However a lower ARI and NMI on Test sets (of custom model) reveal that the higher number was only because of the training docs and instead it degraded the performance of clustering on the test docs (in separating the documents).

Proposed solution (V2):

- **Freeze model internal layers:** While training the embedding model we can only train the last 1-2 layers and keep the rest of the model weights frozen.

Overview (V2): The version 2 performs much better than v1, Focus on ARI and NMI on Test docs which is the best among all.

- V1 do get higher ARI (on all docs) but poor numbers on test docs, which shows overfitting.
- V2 on the other hand shows better separation than baseline throughout (on all docs and also on test docs).

Metrics type and performance:

- Baseline always performed good on internal clustering metrics (which don't require labels). This is based on geometry of the embeddings
- V2 is better on external metrics which need reference labels.
 - This is most likely because the current finetuning of embedding models enforces the objective that chunks from same documents should remain closer and chunks from different documents should be far apart.

held-out documents: ['22-1165_10n2', 'USCOURTS-ilnd-1_04-cv-00397-5']
comparison at k=3, dim=64

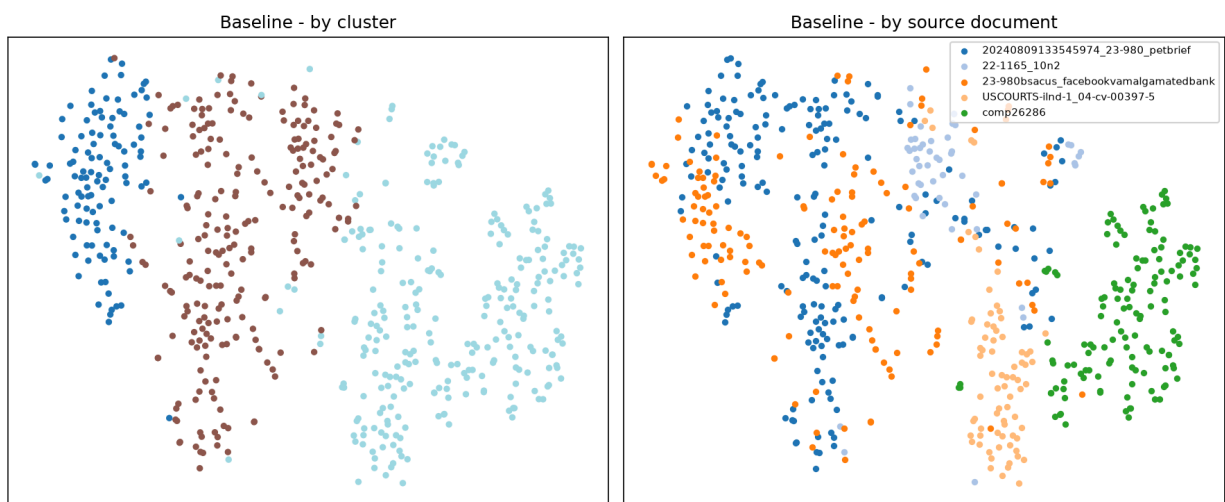
Comparison on Metrics

Metric	Baseline	V1	V2
n_cluster	3	3	3
Silhouette score	0.20	0.18	0.15

Davies Bouldin	2.61	2.68	3.0
Calinski Harabasz	61.34	71.06	55.96
ARI	0.21	0.39	0.35
NMI	0.31	0.52	0.46
Silhouette (on Test Docs)	0.29	0.09	0.25
ARI (on Test Docs)	0.54	0.12	0.56
NMI (on Test Docs)	0.43	0.18	0.48

Clustering plots

Baseline

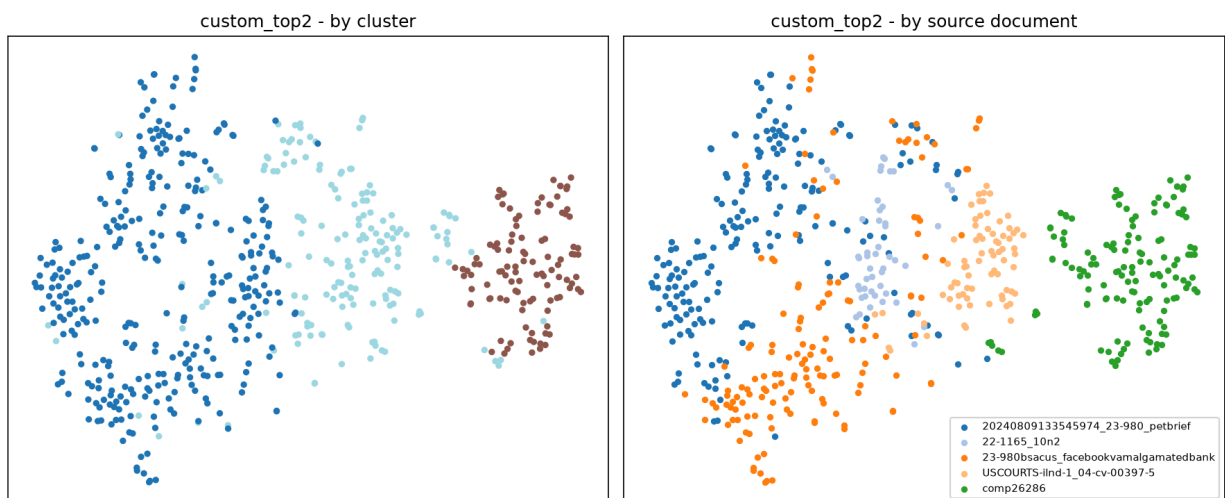


Domain Specific Embedder V1



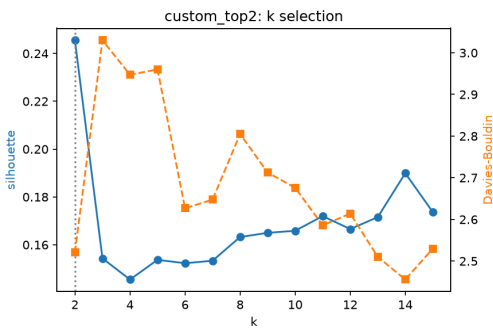
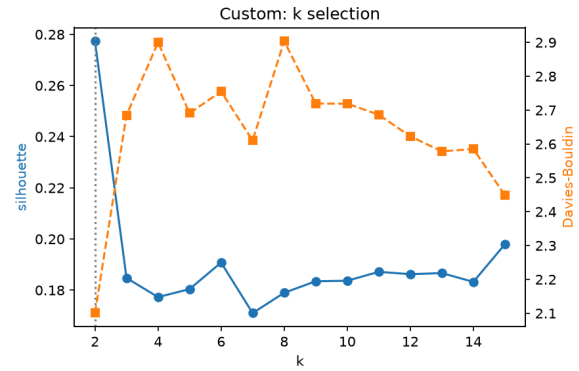
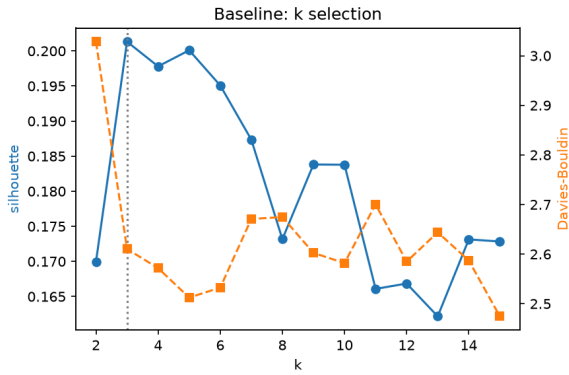
Although we see clear separation among the documents in the training set, the test documents are still not separated clearly (light orange and light blue). Sign of overfitting.

Domain Specific Embedder V2



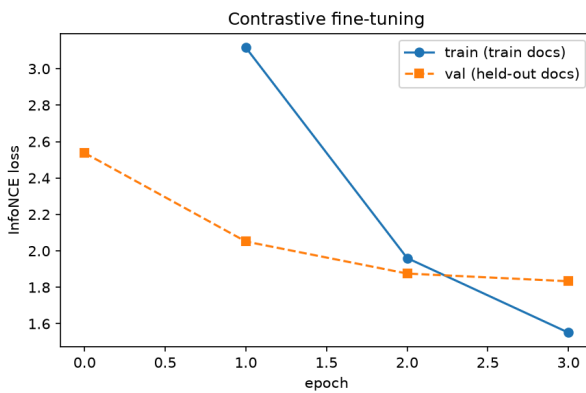
Other param selection

Selecting k for k-mean clustering using silhouette scores



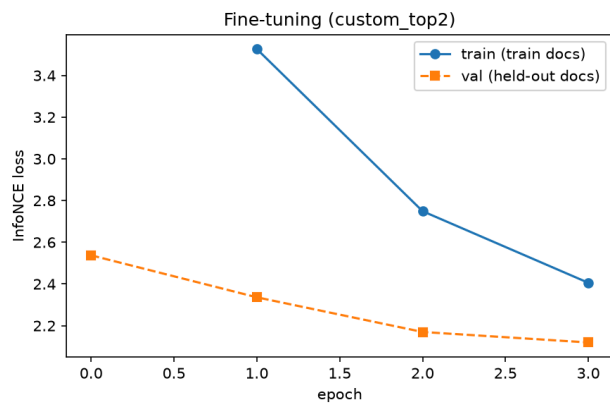
Loss curve of embedding model

V1 Model



The train curve is very sharp so the model was likely overfitting.

V2 Model



A much ideal slope since the slope of validation loss at end is similar to the training loss.